

Web Development Concepts -- Guided Notes

Unit tp-3.10 | CompTIA Network+ N10-009 Obj. FC0-U71 3.10 | N10-008 FC0-U71 3.10

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

HTML stands for _____ and defines the _____ (content and organization) of a web page. CSS stands for _____ and defines the _____ (colors, fonts, layout). JavaScript adds _____ to the page, making it respond to user actions without reloading.

Question 2

HTTP (HyperText Transfer Protocol) is the protocol used to request and serve _____ over the web. HTTPS is the secure version that _____ traffic using TLS. HTTP uses port _____ and HTTPS uses port _____. Browsers display a _____ icon in the address bar for HTTPS sites.

Question 3

DNS (Domain Name System) translates human-readable _____ (like www.google.com) into _____ addresses that computers use to locate servers. Without DNS, users would need to type _____ addresses directly to reach websites. A DNS _____ attack replaces a legitimate IP with a fraudulent one to redirect users.

Question 4

_____ side processing runs on the web server (PHP, Python, Node.js) and sends the finished HTML page to the client. _____ side processing runs in the user's browser (JavaScript) and modifies the page without contacting the server. Modern web apps use both: the server delivers _____ and the client updates the _____ dynamically.

Question 5

A _____ is a pre-built set of code libraries and tools that provides a structure for building web applications. Front-end frameworks include _____ and _____. Back-end frameworks include Django (Python) and _____. Using a framework reduces _____ by providing solved solutions for common tasks.

Question 6

A web browser renders HTML, CSS, and JavaScript into the visual page a user sees. The _____ (Document Object Model) is a tree structure representing the page, which JavaScript can modify. Browsers use a _____ engine to interpret JavaScript (Chrome uses V8, Firefox uses _____). Developer _____ in the browser allow inspecting HTML, CSS, and network requests.

Question 7

A URL (Uniform Resource Locator) has several parts. In the URL `https://shop.example.com/products?id=42`:

`https` is the _____

`shop.example.com` is the _____

`/products` is the _____

`?id=42` is the _____

Question 8 -- Real World Application

REAL WORLD: A student builds a simple website and deploys it using only HTTP (not HTTPS). The site includes a login form where users submit their username and password. A cybersecurity teacher tells them this is dangerous. Explain why HTTP is dangerous for a login form compared to HTTPS, and describe what would happen if an attacker intercepted the traffic.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/tp-3-10-web-development-concepts/tp-3-10-web-development-concepts-answer-key.pdf
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